

Introduction to Data Science — Final Exam Paper 5 (Medium)

Course Code: BCSE2C05 | Semester: II | Max Marks: 100 | Time: 3 Hours Difficulty: ★★☆☆☆ Medium

SECTION A — Short Answer (5 Marks Each, Do Any 4 of 5 = 20 Marks)

Q1. What is Data Normalization? Compare Min-Max and Z-Score normalization.

Data Normalization scales features to a common range so no single feature dominates in distance-based algorithms.

Aspect	Min-Max Normalization	Z-Score (Standardization)
Formula	$(X - X_{min}) / (X_{max} - X_{min})$	$(X - \mu) / \sigma$
Range	[0, 1]	No fixed range (centered at 0)
Sensitive to outliers	Yes (min/max affected by outliers)	Less sensitive
Best for	When data has known bounded range	When data follows normal distribution

Example (Data: 10, 20, 30):

- Min-Max: 10 → 0.0, 20 → 0.5, 30 → 1.0
- Z-Score: $\mu=20, \sigma=8.16 \rightarrow 10 \rightarrow -1.22, 20 \rightarrow 0.0, 30 \rightarrow 1.22$

Q2. Differentiate between Primary and Secondary data with examples.

Aspect	Primary Data	Secondary Data
Definition	Collected first-hand by the researcher.	Pre-existing data collected by someone else.
Source	Surveys, experiments, interviews.	Reports, journals, databases, government publications.
Cost	Expensive and time-consuming.	Cheaper and readily available.
Accuracy	High (specific to research).	May not perfectly fit current need.
Example	Conducting a customer survey for a new product launch.	Using IRCTC booking data to analyse travel patterns.

Q3. Write Python code using NumPy to create an array, reshape it, and compute statistics.

```
import numpy as np

# Create array
arr = np.array([12, 15, 18, 21, 24, 27, 30, 33, 36])

# Reshape to 3x3 matrix
matrix = arr.reshape(3, 3)
print("Matrix:\n", matrix)

# Statistics
print("Mean:", np.mean(arr))           # 24.0
print("Median:", np.median(arr))       # 24.0
print("Std Dev:", np.std(arr))         # 7.48
print("Variance:", np.var(arr))        # 56.0
print("Max:", np.max(arr))             # 36
print("Argmax:", np.argmax(arr))       # Index of max = 8
print("Sum of each column:", np.sum(matrix, axis=0)) # [51, 57, 63]
print("Sum of each row:", np.sum(matrix, axis=1))   # [45, 72, 99]
```

Q4. What are Symmetric and Asymmetric binary attributes? Give examples.

Binary Attributes have exactly two values (0 and 1).

Symmetric: Both values carry equal importance.

- **Example:** Gender (Male/Female) — both values are equally significant.
- Similarity: Both 0-0 and 1-1 matches are counted.

Asymmetric: One value (usually 1) is more important or rarer.

- **Example:** Disease Test (Positive=1, Negative=0) — a positive result is much more significant.
- Similarity: Only 1-1 matches count; 0-0 matches are ignored.

Jaccard Coefficient (for asymmetric binary):

J(A, B) = $f_{11} / (f_{01} + f_{10} + f_{11})$

Example:

	Patient B=1	Patient B=0
Patient A=1	$f_{11} = 2$	$f_{10} = 1$
Patient A=0	$f_{01} = 1$	$f_{00} = 4$

J = $2 / (1 + 1 + 2) = 2/4 = 0.5$

Q5. What are the different Data Collection Strategies? Explain any three.

Strategy	Description	Example
Surveys	Structured questionnaires distributed to respondents.	Google Forms survey on student satisfaction.
Web Scraping	Automated extraction of data from websites.	Scraping Flipkart for product prices using BeautifulSoup.
APIs	Accessing data programmatically from services.	Using Twitter API to collect tweets for sentiment analysis.
Observations	Recording data by watching events.	Counting footfall in a mall at different times.
Experiments	Controlled tests with treatment and control groups.	A/B testing a website's checkout flow.

SECTION B — Analytical Questions (8 Marks Each, Do Any 5 of 6 = 40 Marks)

Q6. Explain the Data Science Lifecycle. Apply it to a real-world scenario of your choice.

Lifecycle Phases applied to "Predicting Student Exam Performance":

Phase	Application
1. Problem Definition	"Can we predict students at risk of failing based on early indicators?"
2. Data Collection	Attendance records, mid-term marks, assignment scores, LMS engagement from university database.
3. Data Cleaning	Fill missing attendance with 0 (assumed absent), remove duplicate entries, standardize date formats.
4. EDA	Histogram of marks distribution → right-skewed. Correlation: attendance and marks = 0.72 (strong positive).
5. Feature Engineering	Create Engagement_Score = (LMS logins × 0.3 + assignments submitted × 0.7).
6. Model Building	Train Logistic Regression (baseline) and Random Forest.

7. Evaluation	Random Forest: Accuracy=88%, Recall=91%, F1=0.87. Better than Logistic (82%).
8. Deployment	Dashboard for faculty showing at-risk students list. Alerts sent when attendance drops below 60%.

Q7. Write Python code to create a complete visualization dashboard with Matplotlib (subplot with 4 charts).

```
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(42)

# Sample data
departments = ['CSE', 'ECE', 'ME', 'CE', 'EE']
avg_salary = [72000, 65000, 58000, 55000, 60000]
ages = np.random.normal(30, 5, 200)
experience = np.random.uniform(1, 15, 50)
salary = experience * 5000 + np.random.normal(0, 5000, 50) + 30000
ratings = np.random.choice(['Poor', 'Average', 'Good', 'Excellent'], 100)

fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle('Employee Analytics Dashboard', fontsize=16, fontweight='bold')

# 1. Bar Chart
axes[0, 0].bar(departments, avg_salary, color=['#2ecc71', '#3498db', '#e74c3c', '#f39c12', '#9b59b6'])
axes[0, 0].set_title('Average Salary by Department')
axes[0, 0].set_ylabel('Salary (₹)')

# 2. Histogram
axes[0, 1].hist(ages, bins=15, color='steelblue', edgecolor='black', alpha=0.7)
axes[0, 1].set_title('Age Distribution')
axes[0, 1].set_xlabel('Age')
axes[0, 1].set_ylabel('Frequency')

# 3. Scatter Plot
axes[1, 0].scatter(experience, salary, alpha=0.6, color='coral', edgecolors='black')
axes[1, 0].set_title('Experience vs Salary')
axes[1, 0].set_xlabel('Experience (Years)')
axes[1, 0].set_ylabel('Salary (₹)')

# 4. Pie Chart
unique, counts = np.unique(ratings, return_counts=True)
axes[1, 1].pie(counts, labels=unique, autopct='%1.1f%%', colors=['#e74c3c', '#f39c12', '#2ecc71', '#3498db'])
axes[1, 1].set_title('Rating Distribution')

plt.tight_layout()
plt.savefig('dashboard.png', dpi=150)
plt.show()
```

Q8. Explain the K-Means algorithm and compute two iterations.

K-Means: Unsupervised clustering that partitions data into K groups.

Given: 2D points: A(1,1), B(2,1), C(4,3), D(5,4), K=2 **Initial centroids:** C1=(1,1), C2=(5,4)

Iteration 1:

Point	d to C1(1,1)	d to C2(5,4)	Cluster
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A(1,1)	$\sqrt{0} = 0$	$\sqrt{(16+9)} = 5$	C1
B(2,1)	$\sqrt{(1+0)} = 1$	$\sqrt{(9+9)} = 4.24$	C1
C(4,3)	$\sqrt{(9+4)} = 3.61$	$\sqrt{(1+1)} = 1.41$	C2
D(5,4)	$\sqrt{(16+9)} = 5$	$\sqrt{0} = 0$	C2

New centroids:

- C1 = ((1+2)/2, (1+1)/2) = **(1.5, 1.0)**
- C2 = ((4+5)/2, (3+4)/2) = **(4.5, 3.5)**

Iteration 2 (C1=1.5,1.0 and C2=4.5,3.5):

Point	d to C1(1.5,1)	d to C2(4.5,3.5)	Cluster
A(1,1)	$\sqrt{(0.25)} = 0.5$	$\sqrt{(12.25+6.25)} = 4.3$	C1
B(2,1)	$\sqrt{(0.25)} = 0.5$	$\sqrt{(6.25+6.25)} = 3.54$	C1
C(4,3)	$\sqrt{(6.25+4)} = 3.2$	$\sqrt{(0.25+0.25)} = 0.71$	C2
D(5,4)	$\sqrt{(12.25+9)} = 4.61$	$\sqrt{(0.25+0.25)} = 0.71$	C2

Same clusters → **Converged!** Final: Cluster1 = {A, B}, Cluster2 = {C, D}

Q9. Explain Data Preprocessing and its steps with code examples for each.

Data Preprocessing transforms raw data into a clean format for analysis.

Step 1: Data Cleaning

```
import pandas as pd
import numpy as np

df = pd.DataFrame({
    'Name': ['Alice', 'Bob', 'Alice', 'Charlie', None],
    'Age': [22, np.nan, 22, 25, 28],
    'Salary': [50000, 60000, 50000, np.nan, 45000]
})

# Remove duplicates
df = df.drop_duplicates()

# Fill missing values
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Salary'] = df['Salary'].fillna(df['Salary'].mean())
df['Name'] = df['Name'].fillna('Unknown')
```

Step 2: Data Transformation

```
# Min-Max Normalization
df['Salary_Normalized'] = (df['Salary'] - df['Salary'].min()) / (df['Salary'].max() - df['Salary'].min())
```

Step 3: Data Integration

```
df2 = pd.DataFrame({'Name': ['Alice', 'Bob'], 'Department': ['CSE', 'ECE']})
merged = pd.merge(df, df2, on='Name', how='left')
```

Step 4: Data Reduction

```
# Feature selection – drop less important columns
df_reduced = df.drop(columns=['Salary_Normalized'])

# Sampling
df_sample = df.sample(frac=0.5, random_state=42)
```

Q10. Compare Histograms, Boxplots, and Scatterplots. When to use each?

Feature	Histogram	Boxplot	Scatterplot
Shows	Frequency distribution	Summary stats (Q1, Q2, Q3, IQR, outliers)	Relationship between 2 variables
Variables	1 numeric	1 numeric (optionally grouped)	2 numeric
Best For	Understanding data shape (normal, skewed)	Comparing distributions, detecting outliers	Finding correlations
Outliers	Not directly visible	Clearly shown as dots	Visible as isolated points
Library	plt.hist()	sns.boxplot()	plt.scatter()

When to Use:

- "What does the salary distribution look like?" → Histogram
- "Are there outliers in marks across departments?" → Boxplot
- "Is there a correlation between study hours and grades?" → Scatterplot

Q11. Explain Ethical Considerations in Data Science with industry examples.

1. Data Privacy & Consent:

- Users must know what data is collected and how it's used.
- **GDPR** mandates explicit opt-in consent.
- **Example:** Aadhaar data breach (India, 2018) — personal details of millions exposed through unsecured APIs.

2. Algorithmic Bias:

- Models trained on biased data produce discriminatory results.
- **Example:** Amazon's AI hiring tool (2018) was biased against women because training data came from 10 years of male-dominated resumes.
- **Fix:** Use balanced datasets, audit models for bias, apply fairness constraints.

3. Transparency & Explainability:

- Black-box models (deep learning) make decisions humans can't understand.
- **Example:** A loan rejection by an AI system without explanation violates right to fair treatment.
- Tools: LIME (Local Interpretable Model-agnostic Explanations), SHAP values.

4. Data Security:

- **Example:** Equifax breach (2017) — 147 million people's Social Security numbers exposed.
- Requires encryption, access controls, regular audits.

5. Responsible AI:

- Consider societal impact of AI applications.
- **Example:** Deepfakes used to create fake videos of politicians — raises serious ethical concerns about misinformation.

SECTION C — Long Answer (20 Marks Each, Do Any 2 of 3 = 40 Marks)

Q12. (a) Describe Descriptive and Predictive Data Mining with real-world examples (10 Marks). (b) Explain the complete Data Science Lifecycle with a diagram (10 Marks).

(a) Descriptive vs Predictive Data Mining:

Descriptive Data Mining — Summarizes data to find patterns.

- **Clustering:** Segment customers into groups (budget, premium, luxury).
- **Association:** "Customers who buy pizza also buy soft drinks" (support=0.65, confidence=0.80).
- **Summarization:** "Average order value increased 15% in Q3."

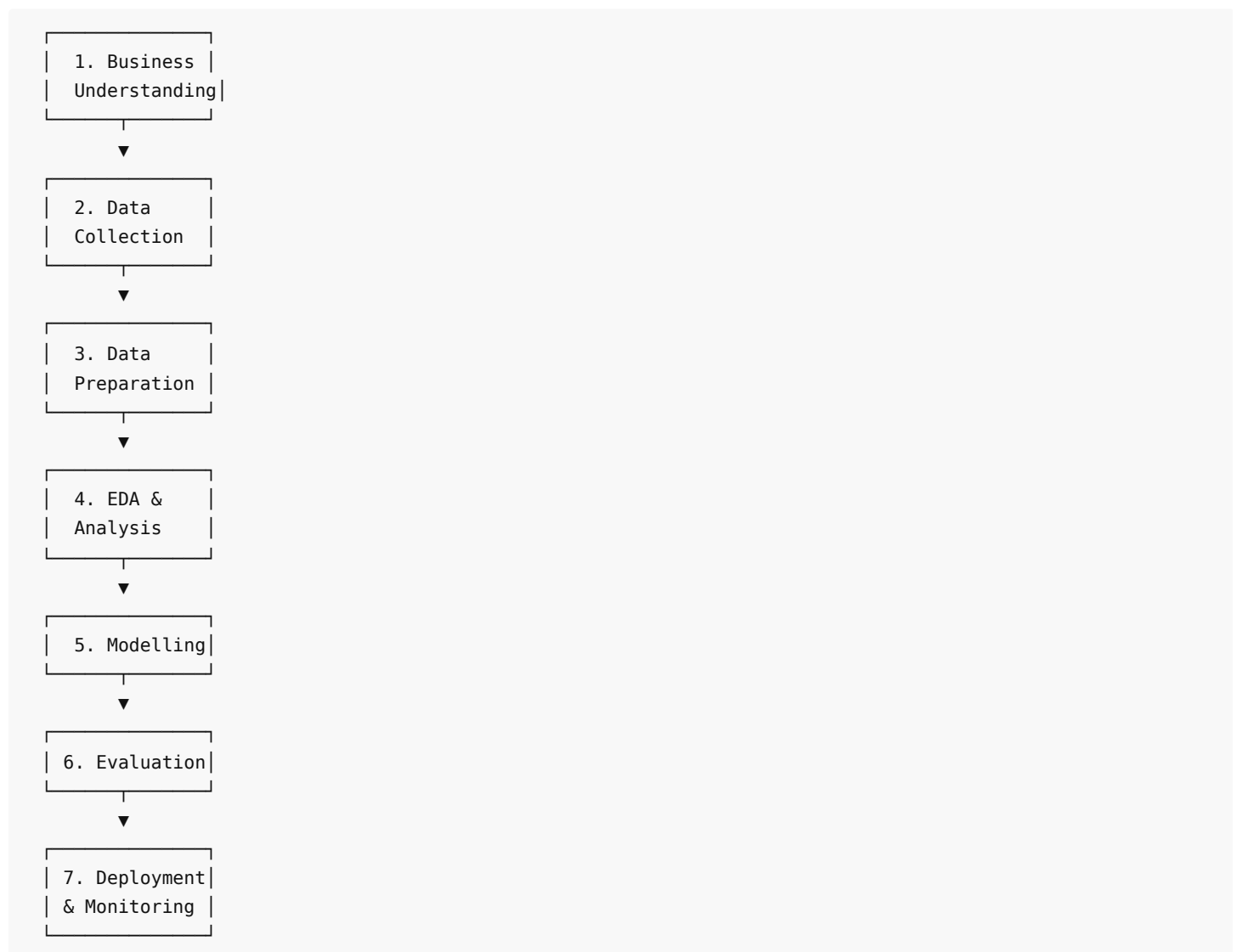
Predictive Data Mining — Uses patterns to forecast.

- **Classification:** Predict if email is spam/not spam.
- **Regression:** Predict house price based on area, location.
- **Time Series:** Predict next month's stock price.

E-commerce Case Study (Flipkart):

- **Descriptive:** "60% of electronics buyers are aged 18-30." "Peak sales happen during Diwali (3x normal)."
- **Predictive:** "User X will likely buy a phone case based on recent phone purchase." "Demand for winter clothing will increase 200% in November."

(b) Data Science Lifecycle:



Each phase feeds back into previous phases when issues arise — it's iterative, not linear.

Q13. A company's HR dataset has: Emp_ID, Name, Age, Gender, Department, Years_Experience, Monthly_Salary, Performance_Rating, Joining_Date, City .Write a complete Python program.

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# — 1. Load Data —
df = pd.read_csv('hr_data.csv')
print("Shape:", df.shape)
print("Dtypes:\n", df.dtypes)
print("First 5:\n", df.head())

# — 2. Descriptive Statistics —
print(f"\nMean Salary: ₹{df['Monthly_Salary'].mean():.2f}")
print(f"Median Experience: {df['Years_Experience'].median()}")
print(f"Mode Gender: {df['Gender'].mode()[0]}")
print(f"Range of Salary: {df['Monthly_Salary'].max() - df['Monthly_Salary'].min()}")
print(f"Variance of Rating: {df['Performance_Rating'].var():.3f}")
print(f"Std Dev of Salary: {df['Monthly_Salary'].std():.2f}")

# — 3. Missing Values —
print("\nMissing:\n", df.isnull().sum())
df['Monthly_Salary'] = df['Monthly_Salary'].fillna(df['Monthly_Salary'].median())
df['Performance_Rating'] = df['Performance_Rating'].fillna(df['Performance_Rating'].mean())
df['Department'] = df['Department'].fillna(df['Department'].mode()[0])

# — 4. Feature Engineering —
def exp_category(y):
    if y < 3: return 'Junior'
    elif y <= 7: return 'Mid-Level'
    else: return 'Senior'

df['Exp_Level'] = df['Years_Experience'].apply(exp_category)
print("\nExperience Level Distribution:")
print(df['Exp_Level'].value_counts())

# — 5. Grouping —
dept_stats = df.groupby('Department').agg({
    'Monthly_Salary': ['mean', 'median', 'std'],
    'Performance_Rating': 'mean'
}).round(2)
print("\nDepartment Stats:\n", dept_stats)

# — 6. Boxplot —
plt.figure(figsize=(10, 6))
sns.boxplot(x='Department', y='Monthly_Salary', data=df, palette='coolwarm')
plt.title('Salary by Department')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# — 7. Histogram + KDE —
plt.figure(figsize=(8, 5))
sns.histplot(df['Performance_Rating'], bins=15, kde=True, color='teal')
plt.title('Performance Rating Distribution')
plt.xlabel('Rating')
plt.show()

# — 8. Scatter —
plt.figure(figsize=(8, 5))
sns.scatterplot(x='Years_Experience', y='Monthly_Salary', hue='Department', data=df, alpha=0.7)

```

```
plt.title('Experience vs Salary')
plt.show()

# — 9. IQR Outliers —
Q1 = df['Monthly_Salary'].quantile(0.25)
Q3 = df['Monthly_Salary'].quantile(0.75)
IQR = Q3 - Q1
lower, upper = Q1 - 1.5 * IQR, Q3 + 1.5 * IQR
outliers = df[(df['Monthly_Salary'] < lower) | (df['Monthly_Salary'] > upper)]
print(f"\nIQR: {IQR:,.0f}, Bounds: [{lower:,.0f}, {upper:,.0f}]")
print(f"Outliers: {len(outliers)}")

# — 10. Save —
df.to_csv('hr_cleaned.csv', index=False)
print("Saved!")
```

Q14. (a) Explain all types of Machine Learning with examples (10 Marks). (b) Describe k-NN and K-Means with manual examples (10 Marks).

(a) Types of Machine Learning:

1. Supervised Learning:

- Labelled data → learns input-output mapping.
- Classification: Spam detection, disease diagnosis.
- Regression: House price prediction, sales forecasting.

2. Unsupervised Learning:

- Unlabelled data → discovers hidden structure.
- Clustering: Customer segmentation.
- Association: Market basket analysis.
- Dimensionality Reduction: PCA.

3. Semi-Supervised Learning:

- Small labelled + large unlabelled data.
- Example: Google Photos — a few labelled face images help identify people across thousands of unlabelled photos.

4. Reinforcement Learning:

- Agent learns through rewards/penalties in an environment.
- Example: Self-driving cars — agent gets positive reward for staying in lane, negative for crossing.

(b) k-NN Manual Example:

Training data:

Point	X	Y	Class
A	2	3	Cat
B	3	4	Cat
C	6	7	Dog
D	7	8	Dog

New point P(4, 5), k=3:

- $d(P,A) = \sqrt{(4-2)^2 + (5-3)^2} = 2.83$
- $d(P,B) = \sqrt{(4-3)^2 + (5-4)^2} = 1.41$
- $d(P,C) = \sqrt{(4-6)^2 + (5-7)^2} = 2.83$
- $d(P,D) = \sqrt{(4-7)^2 + (5-8)^2} = 4.24$

3 nearest: B(Cat,1.41), A(Cat,2.83), C(Dog,2.83) → **2 Cat, 1 Dog** → **Cat**

(c) K-Means Manual Example:

Data: {2, 3, 8, 9, 10}, K=2, C1=2, C2=10

Point	d to C1=2	d to C2=10	Cluster
2	0	8	C1
3	1	7	C1
8	6	2	C2
9	7	1	C2
10	8	0	C2

New: C1=(2+3)/2=**2.5**, C2=(8+9+10)/3=**9.0**

Iteration 2 with C1=2.5, C2=9.0 → same assignments → **Converged!**